CS-1101
PROGRAMMING FUNDAMENTALS
(2+1)

Lab 01

By

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October 13 & 14

Schedule

- Lectures & Labs:

  16 Lectures + 14 Labs

- Length :

  2 hrs / week Lecture

  3 hrs / week Lab

- Day/Time :

  Tuesday 10:00 AM – 12:00 PM (Lecture)
  Monday 10:00 AM – 01:00 PM (Lab CET)
  Monday 01:00 PM – 04:00 PM (Lab SE)

- Office Hrs :

  Wed 11:00 AM – 02:00 PM

Contents

- Introduction to Programming and languages
- Algorithms, Flowcharts and Pseudo code
- Variables and data types
- Operators and expressions
- Control Structures
- Arrays
- Functions
- String and String Operations
- Pointers
- File I/O Operations

Textbooks

#	Title	Author
01	C++ Programming: From Problem Analysis to Program Design, 6th Edition	D.S. Malik
02	Object Oriented Programming in C++ , latest edition.	Robert Lafore
03	C How to Program, latest Edition	Paul Deitel & Harvey Deitel
04	The C Programming Language, latest Edition	Brian W. Kernighan, Dennis M.Ritchie

Learning Outcomes

CLO	Statement	Learning Levels	PLO
CLO 1	Discuss the basic programming concepts.	*C-2 Comprehension (Level 2)*	PLO 1 *Engineering Knowledge*
CLO 2	Apply programming techniques to solve various problems.	C-3 *Application (Level 3)*	PLO 3 *Design/Development of Solution*
CLO 3 LAB	Apply fundamental programming concepts to solve computational problems.	C-3 *Application (Level 3)*	PLO 3 *Design/Development of Solution*

Individual score % to pass CLO = 50 %
Cohort score % to pass CLO = 60%

Evaluation

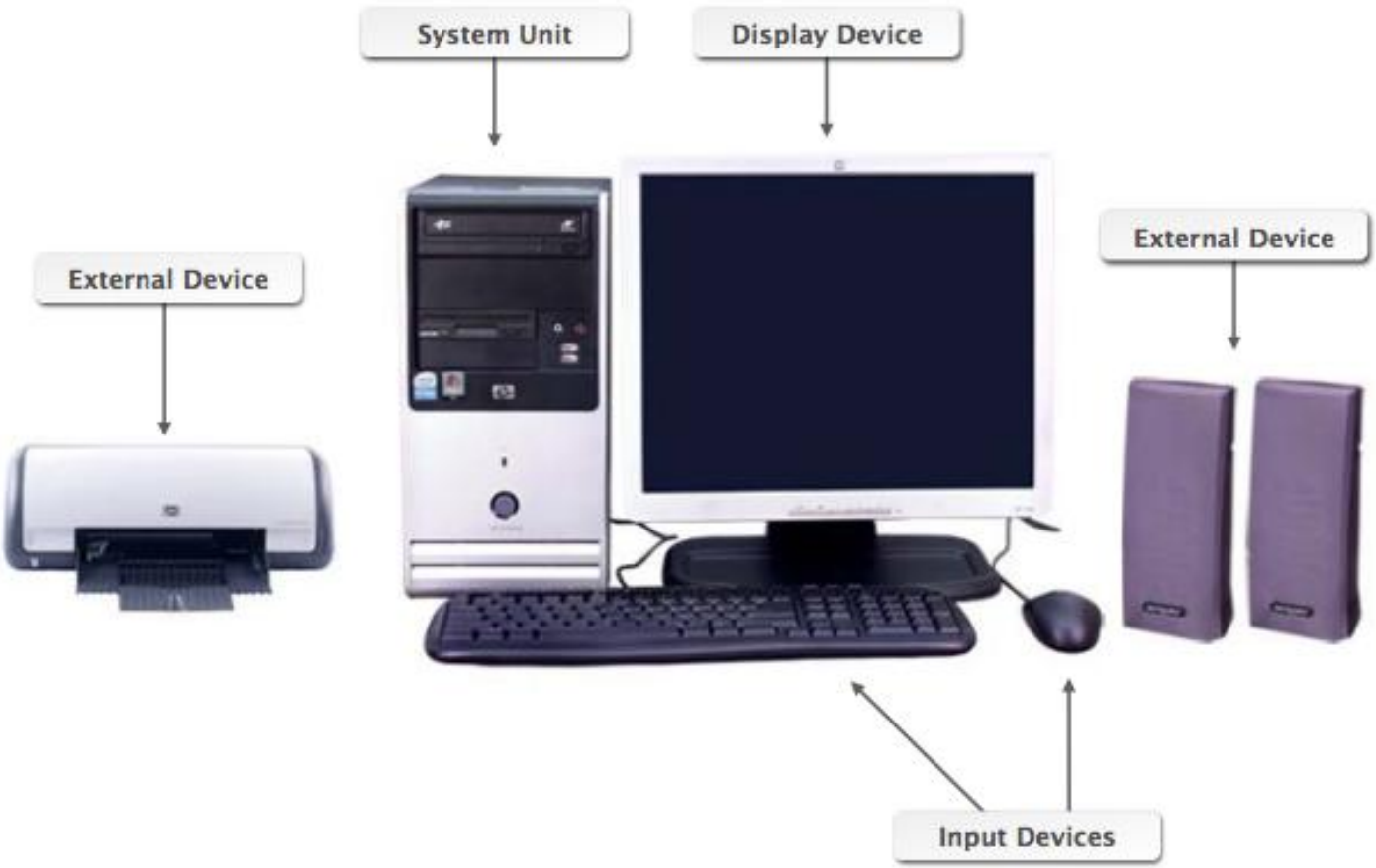
Activity	Qty
Quiz	Min 2
Assignment	Min 2
Midterm	01
Final Term	01
Design Project + Presentation	01
Lab Tasks	13
Open Ended Lab	01

PROGRAMMING FUNDAMENTALS

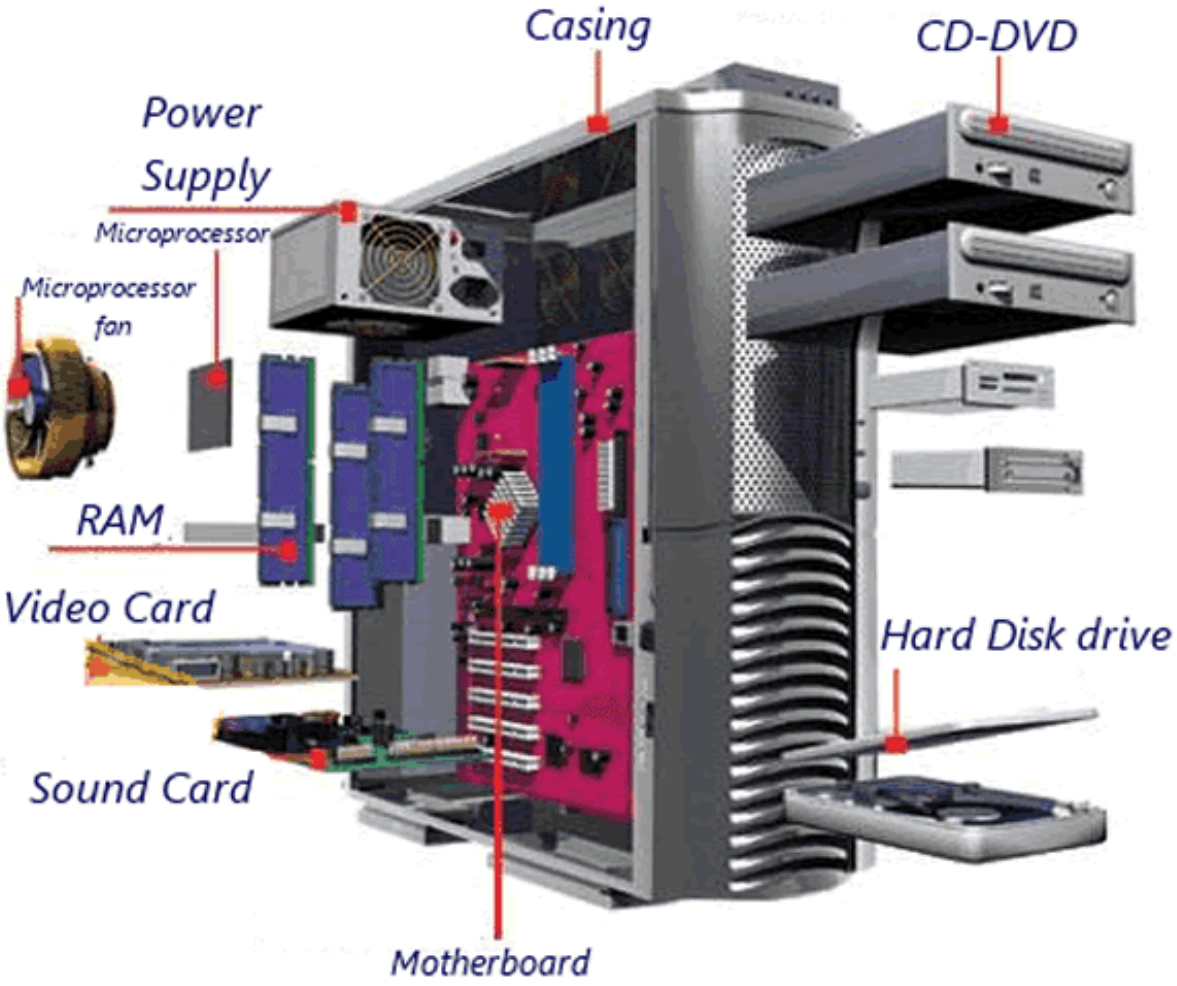
WHAT IS A COMPUTER ?

- A computer is a programmable electronic device that accepts raw data as input and processes it with a set of instructions (a program) to produce the result as output.
- Two main components of computer are:
  - Hardware
  - Software

HARDWARE


System Unit
Display Device
External Device
External Device
Input Devices

HARDWARE cont..



SOFTWARE

- Software are programs that enable the user to interact with a computer, its hardware, or perform tasks.
- Without software, computers would be useless.
- Types of Software :
  - System Software
  - Application Software

SOFTWARE cont…

- System software are programs designed to run a computer's hardware and provides a platform for applications to run on top of.
- Application software are programs designed to perform a specific tasks.

SOFTWARE cont…

COMPUTER PROGRAM

- A computer program is a sequence of instructions written using a Computer Programming Language to perform a specified task by the computer.

Consider a situation when someone asks you about how to go to a nearby Petrol Pump. What exactly do you do to tell him the way to go to Petrol Pump ?

First go to the University main gate , take a left turn and then drive straight for half kilometer and you will find Petrol pump at the right.

You have used English Language to give several steps to be taken to reach Petrol Pump. If they are followed in the following sequence, then you will reach Petrol pump.

PROGRAM

1. Go to University Main Gate
2. Take a left turn
3. Drive straight for half kilometer
4. Search for petrol pump on your right side.

Human Program written in English Language

Can we write the above Petrol Pump Program in any other language like Urdu, French, Spanish or any other human language etc..?

YES

provided the person seeking direction knows any of these languages.

COMPUTER PROGRAM

```
#include<iostream>
using namespace std;
int main(){
    int number1 = 10;
    int number2 = 20;
    int sum = number1 + number2;
    cout << sum << endl;
    return 0;
}
```

Sample Run:

30

COMPUTER PROGRAMMING AND LANGUAGES

- The act of writing computer programs is called computer programming.
- There are hundreds of programming languages, which can be used to write computer programs.
  - Java
  - C
  - C++
  - Python
  - PHP

The Evolution of Programming Languages

- Low-level Language
  - Machine Language
  - Assembly Language
- High-level languages include Basic, FORTRAN, COBOL, Pascal, C, C++, C#, and Java.

TRANSLATORS

- **Assembler:** translates a program written in a assembly language to machine language.
- **Compiler:** translates a program written in a high-level language to machine language.

Interpreter ?

Processing a C++ Program

```
#include <iostream>
using namespace std;
int main()
{
    cout << "My first C++ program." << endl;
    return 0;
}
```

Sample Run:

```
My first C++ program.
```

Processing a C++ Program (cont'd.)

- To execute a C++ program:
  - Use an editor to create a source program in C++
  - Preprocessor directives begin with # and are processed by the preprocessor
  - Use the compiler to:
    - Check that the program obeys the rules
    - Translate into machine language (object program)

Processing a C++ Program (cont'd.)

- To execute a C++ program (cont'd.):
  - Linker:
    - Combines object program with other programs provided by the SDK to create executable code
  - Loader:
    - Loads executable program into main memory
  - The last step is to execute the program

Processing a C++ Program (cont'd.)

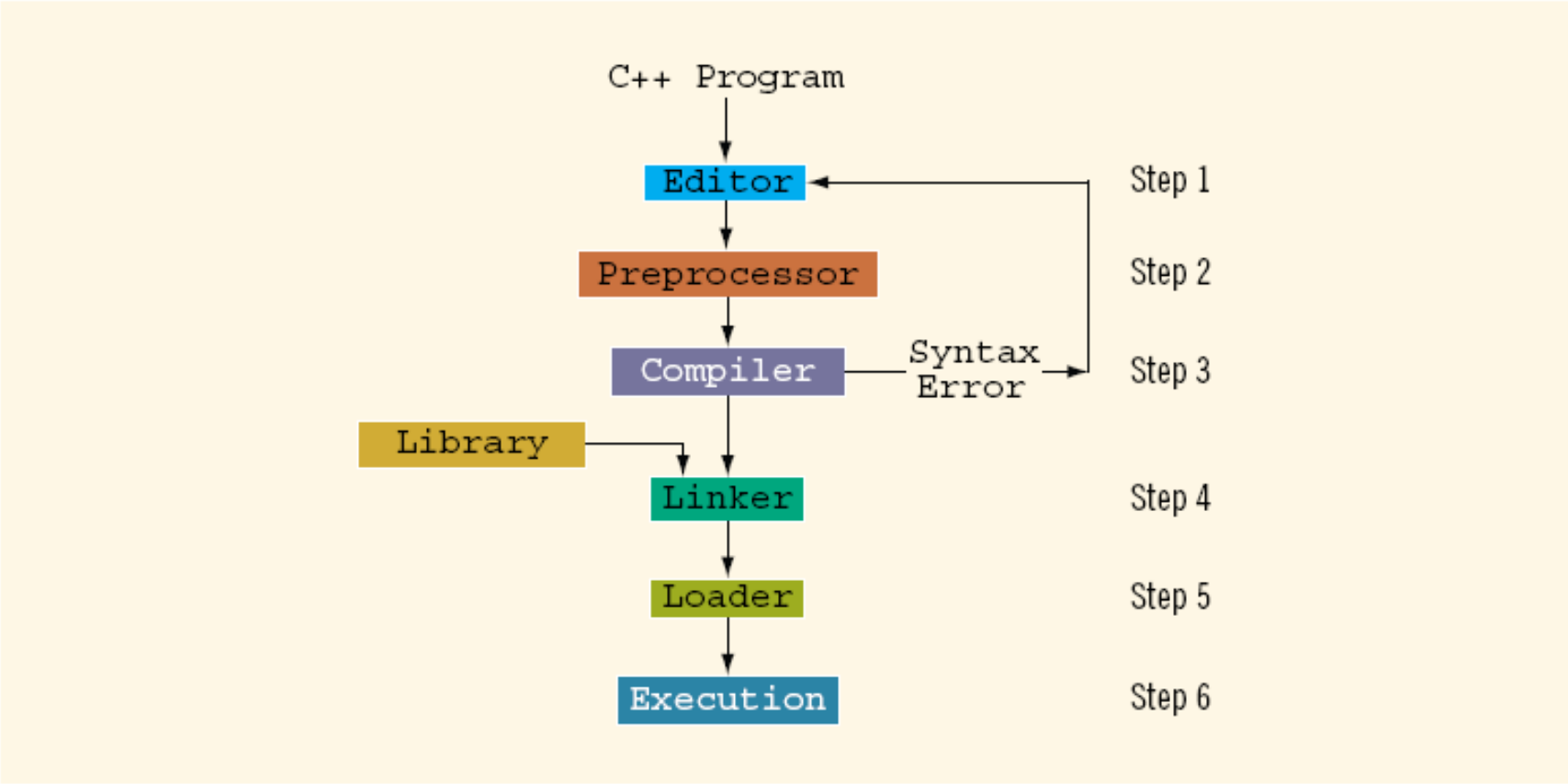


FIGURE 1-3 Processing a C++ program

Programming with the Problem Analysis–Coding–Execution Cycle

- Programming is a process of problem solving
- One problem-solving technique:
  - Analyze the problem
  - Outline the problem requirements
  - Design steps (algorithm) to solve the problem
- <u>Algorithm</u>:
  - Step-by-step problem-solving process
  - Solution achieved in finite amount of time

The Problem Analysis–Coding–Execution Cycle (cont'd.)

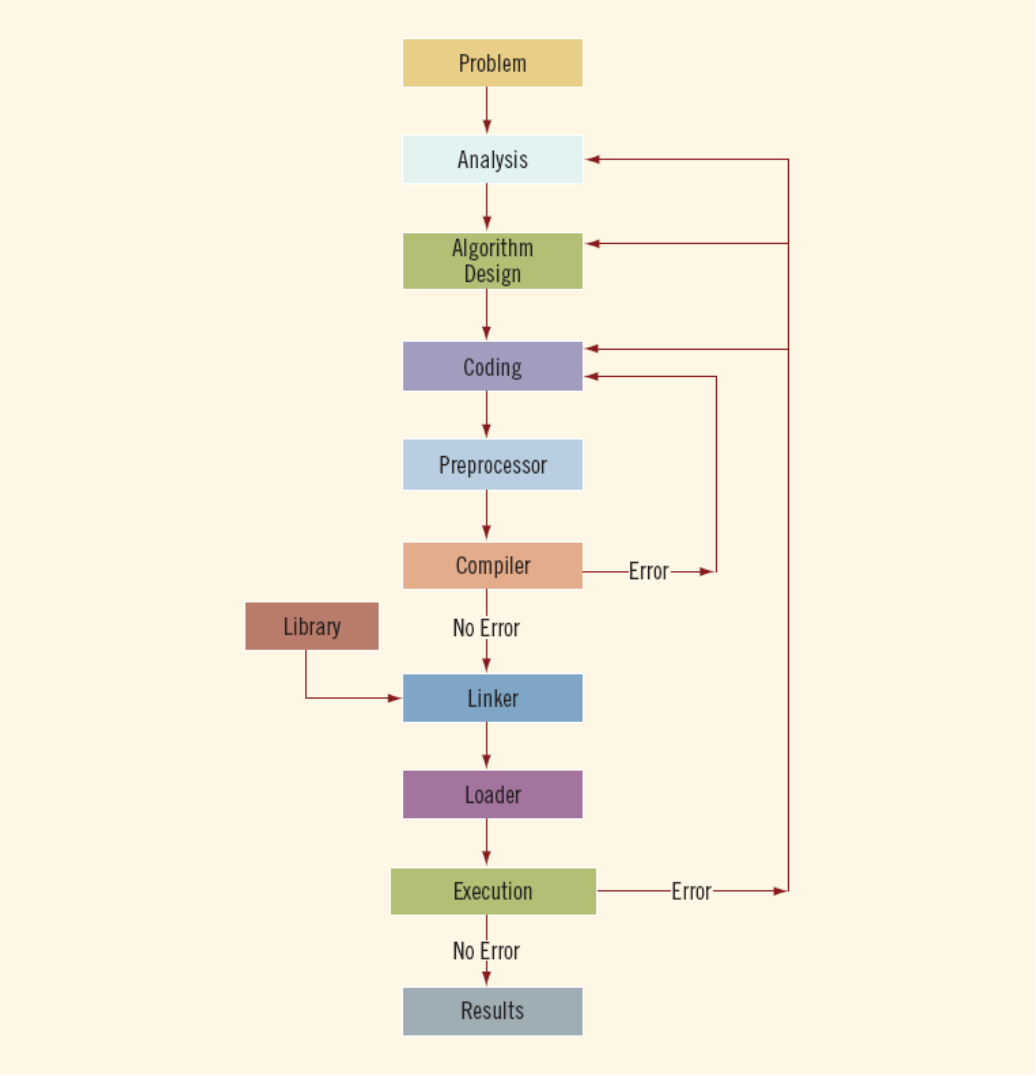


FIGURE 1-4 Problem analysis–coding–execution cycle

The Problem Analysis–Coding–Execution Cycle (cont'd.)

- Run code through compiler
- If compiler generates errors
  - Look at code and remove errors
  - Run code again through compiler
- If there are no syntax errors
  - Compiler generates equivalent machine code
- Linker links machine code with system resources

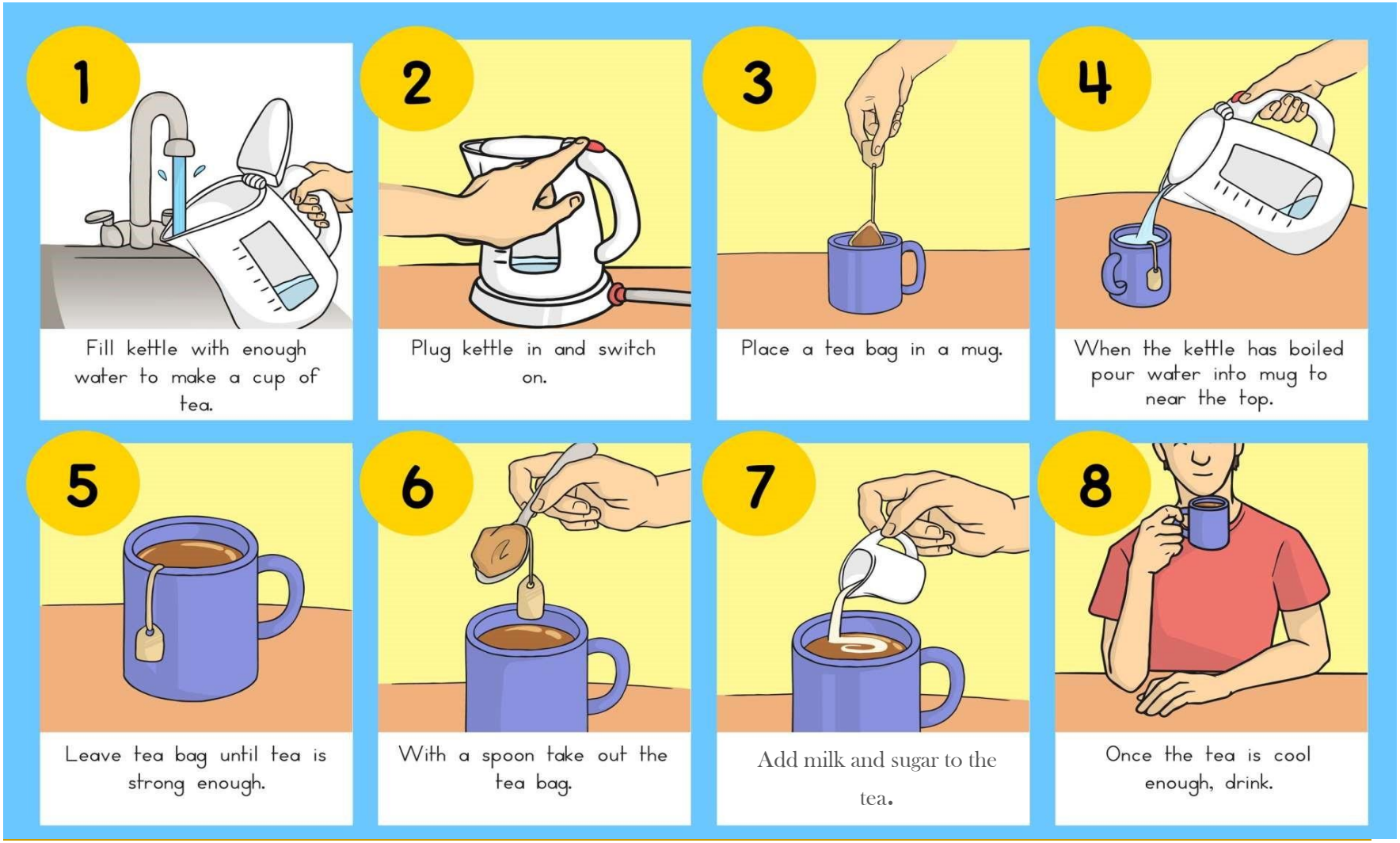
The Problem Analysis–Coding–Execution Cycle (cont'd.)

- Once compiled and linked, loader can place program into main memory for execution
- The final step is to execute the program
- Compiler guarantees that the program follows the rules of the language
  - Does not guarantee that the program will run correctly

Algorithm

CAN YOU WRITE STEPS TO MAKE A CUP OF TEA?

Algorithm


1
Fill kettle with enough water to make a cup of tea.
2
Plug kettle in and switch on.
3
Place a tea bag in a mug.
4
When the kettle has boiled pour water into mug to near the top.
5
Leave tea bag until tea is strong enough.
6
With a spoon take out the tea bag.
7
Add milk and sugar to the tea.
8
Once the tea is cool enough, drink.

Algorithm:

- **Step-1** Start
- **Step-2** Fill Kettle with enough water to make a cup of tea.
- **Step-3** Plug Kettle in and switch on.
- **Setp-4** Place a tea bag in a mug.
- **Setp-5** When the water has boiled pour water into mug near to top .
- **Setp-6** Leave a bag in a mug until tea is strong enough.
- **Setp-7** With a spoon takeout the teabag.

- **Step-8** Add milk and sugar to the tea.
- **Setp-9** Once the tea is cool enough , drink it.
- **Setp-10** Stop

Example 1-1

- Design an algorithm to find the perimeter and area of a rectangle
- The perimeter and area of the rectangle are given by the following formulas:

```
perimeter = 2 * (length + width)
area = length * width
```

Example 1-1 (cont'd.)

- Algorithm:
  - **Step-1** Start
  - **Step-2** Get length of the rectangle
  - **Step-3** Get width of the rectangle
  - **Setp-4** Find the perimeter using the following equation:

```
perimeter = 2 * (length + width)
```

  - **Step-6** Find the area using the following equation:

```
area = length * width
```

  - **Step-7** Stop

Escape sequences

- Escape sequences are special characters preceded by a backslash **(\)** and are used to perform specific actions, like inserting a *new line, tab, or double quote* inside a string.
- Escape sequences are commonly used to format output.

Commonly used escape sequences

Escape Sequence's	Meaning	Explanation	Example Usage	Example Output
\n	Newline	Moves the cursor to the next line, creating a line break.	cout << "Hello\nWorld";	Hello World
\t	Horizontal Tab	Adds a horizontal tab space, useful for aligning text in columns.	cout << "Name:\tJohn";	Name: John
\\	Backslash	Prints a single backslash character (\), often necessary in file paths.	cout << "Path: C:\\Files";	Path: C:\Files
\"	Double Quote	Allows double quotes to appear within a string, useful in dialogue or quotations.	cout << "\"Hello!\"";	"Hello!"
\'	Single Quote	Enables single quotes within a string, useful for contractions or quotations.	cout << "It\'s fine";	It's fine
\a	Alert (Beep)	Produces a beep sound on some systems, often used to signal the end of a process.	cout << "\a";	(Triggers a beep sound, if supported)
\r	Carriage Return	Moves the cursor to the beginning of the current line, allowing text to be overwritten.	cout << "Hello\rWorld";	World
\b	Backspace	Moves the cursor one position backward, effectively deleting the previous character.	cout << "Hello\b!";	Hell!

TASK

- Download and install Dev C++ & get yourself familiarize with editor, compiler by writing simple program that prints your name on the screen.

TASK – Cout

Objective: Use cout to display an output on the console.

- Task 1: **Simple Welcome Message**
  - Write a program that outputs a three-line welcome message using separate **cout** statements.
  - Each line should be displayed on a new line using endl.

```
D:\Data\Awais_Laptop\Univer
Welcome to the C++ Lab!
You will learn basic output today.
Let's start with simple messages.

--------------------------------
```

- Task 2: **Printing a Pattern**
  - Write a program that outputs the following pattern on the console.

```
D:\Data\Awais_Laptop\Univer
*
* *
* * *
* * * *
* * * * *
```

TASK – Escape Sequence's

Objective: Use escape sequences in C++ for formatting console output, enhancing readability and presentation.

- Task 1: **Formatting with tabs and newline character.**
  - Write a program to display text in a tabular format using \t & \n.
  - The output should have three columns with labels on one line and sample data on the next.

```
D:\Data\Awais_Laptop\Univer

Item     Price    Quality
Pen      $1       10
Pencil   $2       5
Eraser   $0.5     20

--------------------------------
```

- Task 2: **Message using a variety of escape sequences.**
  - Write a program that displays a short message containing quotes, a file path with backslashes, and a newline.

```
D:\Data\Awais_Laptop\Univer

She said, "Let's learn C++!"
To start, save your file in C:\C++_Projects\Lab1
Good luck!

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```

REFERENCES

- Computer Programming - Overview (tutorialspoint.com)
- Chapter 1: An Overview of Computers and Programming Languages, C++ Programming: From Problem Analysis to Program Design, Fifth Edition, DS Malik